

Worksheet 4 (WS4) – Partitioning Around Medoids (PAM)

Suppose we measure two variables X_1 and X_2 for each of four items A, B, C and D. The data are given in the following table

Item	Observations	
	x_1	x_2
A	5	3
B	-1	1
C	-4	4
D	-3	-2

- (a) Write down lines of R code that will divide these items into 2 clusters using PAM (Partitioning Around Medoids). Use Manhattan distance as the dissimilarity measure.
- (b) Compare the answer you obtain from the R output in part (a) to the results using K-means (Worksheet 2). For this problem, are the results the same or different?
- (c) Explain how to use the silhouette method to estimate the optimal number of clusters in PAM.